Aschaffenburg District Office
- District Fire Inspection -

Basics
the
Vegetation fire fighting
for the
Fire departments

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Contents

1. Introduction

One of the key findings in forest firefighting is that firefighting teams on the ground using manual extinguishing techniques in addition to TLF and extinguishing water drops are extremely effective for lasting operational success.

Firefighting crews fight ground fires, apply wound strips, carry out follow-up extinguishing operations and prevent the fire from spreading even without the support of fire engines.



To enable firefighting teams to fight vegetation fires independently, efficiently, and autonomously, the Aschaffenburg district maintains appropriate vegetation fire modules. These are described in detail in the "Equipment" document. If necessary, these modules can be transported to the respective scene with ease from a logistical perspective.


The strength of an independent firefighting team should be at least one group.

2. Self-sufficient firefighting team

At first glance, firefighting crews as a self-sufficient means of action seem rather helpless and ineffective. However, in many forest fires around the world, they bear the brunt of the firefighting effort—and successfully.

Self-sufficient firefighting teams cut firebreaks at suitable locations, use fire as a means of securing these firebreaks if necessary, and monitor large areas for possible fire islands, which can mainly be caused by flying sparks.

When fighting vegetation fires with self-sufficient firefighting teams, the basic principle is that fires are not extinguished, but primarily contained and controlled. To this end, the self-sufficient units use natural firebreaks and/or create firebreaks, taking particular account of forest fire behavior, vegetation, and topography, and thus contain the fire (see section 14.3). In doing so, they must identify and avoid particularly dangerous areas (such as hillside roads, steep slopes, converging valleys, etc.).

Conclusion: A self-sufficient firefighting team can be used not only to contain a vegetation fire, but also to protect objects, for example residential areas or individual buildings.

3. Equipment for a group

Various lectures, teaching materials, and essays on vegetation fires have demonstrated that the use of appropriate hand tools can achieve significant and visible results in firefighting. This is especially true where extinguishing water is available in limited quantities. Reasons for this may include, for example, a lack of access roads in large forested areas or a lack of tankers and fire engines (e.g., due to multiple simultaneous fires in one region).

In order to ensure that fire fighting can be carried out effectively and independently, the following hand tools and equipment have been proven to be useful for a firefighting team in
Group strength has proven itself and should generally be available to the emergency services stand:

• 2 forest fire/fire beaters
• 2 US forestry shovels • 2 water backpacks/
backpack sprayers • 1 chainsaw with accessories (such as cut-resistant leg warmers, head protection, tools,
Spare chain, canister with fuel and saw chain oil)

The following tools should also be available for applying wound strips and extinguishing the fire: • 2 flat axes/forest fire axes (e.g. Pulaski axe)

• 2 pieces of forest fire universal tool “GORGUI Master”

4. Distribution of equipment to a firefighting group

Depending on the aforementioned purpose, the firefighting team (firefighting group) is assembled and equipped with the following equipment:

Group leader (GF)ÿ heads the E-position

Machinist (Ma) **ÿ** chainsaw or 1 water backpack

Reporter (Me) **ÿ** clear away the cut material or 1 fire beater

Attack Squad (AT) **ÿ** 1 Fire Beater / 1 Sand Shovel
• to support a fire-fighting operation

Water Crew (WT) **ÿ** 1 water backpack / 1 Pulaski
• to combat fire islands or fire nests

Hose team (ST) **ÿ** 1 forest fire universal tool / 1 flat hoe
• for applying wound strips

The distribution of equipment among the individual squads is only an example. The explicit allocation of equipment, or which and how many equipment will be used, is carried out on the instructions of the unit or group leader.

5. Protective clothing for vegetation fire fighting

Personal protective equipment is essential when firefighting teams are deployed. A balance must be struck between the optimal protection against short-term Flame contact, flying sparks, and the highest possible level of comfort. Multi-layered outerwear (fire protection clothing, e.g., according to HuPF or EN 469) places particular strain on the wearer in warm weather. However, working without outerwear is not recommended, as even slight flying sparks can cause injuries. Ideally, long underwear and a single-layer turnout suit are recommended. However, each fire department must determine whether the protective suits available, e.g., Bayern 2000 or Oberstdorf, provide sufficient heat protection.

To facilitate work in slightly smoky areas, a vegetation fire protection mask (Hot Shield) with an integrated particle filter and tight-fitting, fog-free safety goggles should be worn. These not only protect exposed skin areas, such as the neck, throat, and large parts of the face, from flames, flying sparks, and thermal radiation, but also protect the respiratory tract from dust and soot particles thanks to the insertable particle filter. The goggles also protect the firefighter from flying sparks, smoke, and thermal radiation.

Exposed skin areas (face, neck, and throat) can also be protected by a flame-retardant hood or neck protector, or, for longer operations, by a lightweight helmet with a helmet scarf. Respiratory masks, on the other hand, should only be used in extreme Conditions in densely smoke-filled areas. In case of escape from the fire zone, a respiratory mask with a filter must be worn.
Protective gloves are also part of personal equipment. Sunscreen and insect repellent should also be available to emergency personnel.
When using protective shoes, it is recommended, especially in rough terrain Wearing sturdy lace-up boots in the forest reduces the risk of twisting or injuring yourself because the foot is snugly enclosed. Since firefighting involves walking on burned surfaces and embers, the soles of the shoes must be heat-resistant and adequately secured.

6. Additional equipment:

6.1 Warning device

In order to be able to warn each other in the event of sudden dangers, every emergency worker should be equipped with a signal whistle.

6.2 Drinking bottles

When performing physical work near fire, there is a risk of heat exhaustion.

Therefore, you must carry enough liquid with you.
For this purpose, a corresponding number of stainless steel drinking bottles with a capacity of 0.9 liters are available.

They are equipped with a thermal cover and stored in standard beverage crates. The drinking bottles are kept at the catering facility (Holz Volunteer Fire Department).

6.3 Thermohüllen (Thermo Bottle Bag)

The thermal sleeve is a highly insulating drinking bottle bag. It keeps drinks hot or cold for longer. Additionally, the thermal sleeve protects drinking bottles from dirt, dust, and dents. The thermal sleeve is made of polyester fabric with insulating foam. The thermal bottle bag is also equipped with an adjustable belt loop and a tank cap.

The case can hold aluminum or stainless steel drinking bottles with a maximum capacity of 1.5 liters or PET bottles with a maximum capacity of 1 liter.

7. Catering

Due to the heavy physical strain involved in vegetation fire operations, it is important to ensure that emergency personnel consume enough fluids. They should drink plenty of fluids before and during operations to prevent dehydration.

Fruit juice spritzers or water are best suited for this. Apple spritzer has proven particularly effective.

Rule of thumb: *Each emergency worker should drink one liter of liquid (water) per hour, spread over time.*

Likewise, a sufficient energy supply in the form of fruit, bread and sausage (chocolate bars only in exceptional cases) is immensely important during the operation.

8. Personnel (firefighting team) and personnel replacement

Heavy equipment and helicopters are used to fight vegetation fires, especially forest fires. Firefighting crews are tasked with extinguishing ground fires (offensive fighting) and establishing firebreaks and fire lanes (defensive fighting). They also carry out follow-up extinguishing operations and prevent the fire from spreading through flying sparks, even without the support of fire engines. This largely involves strenuous physical labor. To protect and relieve the firefighting crews, sufficient firefighting and operational personnel must be available as backup, and these personnel must be replaced regularly.

The smooth interaction of the individual actors, as well as sufficient firefighting teams (firefighting personnel) capable of fighting the fire, ultimately determine whether the fire is fought effectively or whether the fire is extinguished quickly.

9. Safety rules

The following basic safety rules, which are based on operational and training experience, should be known to all emergency personnel.

ÿ B (S)– Observation / security post

An independent observer should be stationed at each operational section to warn deployed personnel in the event of sudden changes in the weather (wind) or the appearance of hot spots. To ensure rapid evacuation of the endangered area even in the event of a change in the situation under difficult conditions, a clear retreat signal should be agreed upon for all deployed emergency personnel. The observation post must not be involved in firefighting measures or other work, so that the tasks described above can be carried out responsibly and without restrictions. Furthermore, their task is to ensure communication with the section management/operational command and to relay important information.

ÿ K– Ensure communication

Ensuring contact with the command center and ensuring communication within the unit are extremely important during vegetation fires. If this cannot be ensured, there is a very high risk that emergency personnel could be surprised by sudden changes in the situation and thus potentially endangered.

ÿ R– Ensure return/escape route

Each operational area should have a suitable, pre-determined escape route.
This route must be low-fire and familiar to all emergency personnel.
Escape/rescue routes can be marked using tape (small strips tied to trees or bushes) and should lead to a safety zone.

ÿ S– Set up security zone

A safety zone is an area in which emergency responders are safe from fire without additional protective measures, and where the fire's edge does not pose a threat to emergency responders. This can be a road, a parking lot, a rocky area, a large forest clearing, or a body of water. If necessary, this zone must be created by removing vegetation. In forest stands, a safety zone should be maintained that is at least one tree length, or preferably twice the length, of the adjacent vegetation.

10. Firefighting with fire engines

Ground-based firefighting teams are usually supported by fire engines supported. In principle, only off-road vehicles should be used.
The most important features for operations in rough terrain are all-wheel drive, the shortest possible vehicle, ground clearance, a low total weight and low ground pressure, and suitable tires (with adjustable air pressure if necessary). For off-road operations, vehicles must be equipped with anti-skid chains as additional equipment. For fighting vegetation fires, additional equipment must be included in addition to the minimum equipment for the existing vehicle load. This includes additional water-carrying fittings and hand tools (as listed in Section XX) for at least one firefighting group.

It is advantageous if the fire engines have a "Pump & Roll function" with which allows the release of extinguishing water while driving. In the early stages of forest and wildfires, the available water quantities must be used extremely efficiently. Especially when using mobile water tanks, which have a limited capacity, the optimal application rate must be coordinated. The addition of wetting agents (environmental compatibility should be taken into account) can reduce the effectiveness and the usually prevailing shortage

of extinguishing agents. Air, compressed air foam, and gelling agents are suitable for securing objects and firebreaks. Since most forest fires are ground fires, nozzles with a spray jet are primarily used.
The advantage here is the particularly good maneuverability even in difficult terrain (e.g. D-hoses and D-nozzles). In areas where open flames and glowing components endanger the use of conventional hoses, rubberized or water-permeable hoses can be used.

Only after the outer perimeter of the fire has been extinguished can the fire engine begin extinguishing the entire fire area. Before deploying the fire engine, care must be taken to ensure that the intended fire location is completely extinguished. This prevents damage to the control and supply lines.

During the operation, the vehicles must be positioned so that escape is possible if the direction of fire changes. This means reversing to the scene (see image) so that escape can be made quickly in case of danger.
Care must be taken to ensure that parked vehicles do not impede the access of advancing units.

Reverse to the scene.

Keep the paths clear!

In the case of vegetation fires, oncoming traffic should generally be avoided in the access and exit areas of staging areas, water intake points, and emergency sites. It is recommended to establish a one-way traffic system for these areas.

11. Firefighting from the air

Aerial forest firefighting can fundamentally only be viewed as a supporting measure for ground-based firefighting. Nevertheless, it is essential for large wildfires and often represents the only means of fighting them. Large-scale water drops from aircraft are primarily used to combat rapidly spreading fires. Even munitions-contaminated and difficult-to-access areas, such as alpine terrain, can be reached with air support.

Localized water drops are used for targeted follow-up extinguishing. In Germany, helicopters used primarily by the police, the German Armed Forces, or private service providers (e.g., www.helialert.com) with special external load containers (ALB) are used. In addition to helicopter transport of extinguishing water, the transport of personnel and equipment also plays an important role in Alpine regions.

A clear operational strategy should be developed before the helicopter is deployed.
It must be determined where helicopters can refuel their ALBs and where they can specifically attack the fires. The plan must also indicate where vehicles can be refueled and parked for breaks.

To avoid endangering ground-based emergency personnel, it is necessary to clearly define in advance the areas in which water can be dropped from the air. During aerial firefighting, the target area must be cleared by firefighting crews.

If firefighting crews remain in the target area, they should avoid getting wet if possible. The risk of falling branches or, on slopes with rolling stones, being hit by helicopters is also increased.

When fighting fires from the air, colored extinguishing agent additives
which serve to mark the treated area.

Use of a helicopter with external fire water tank

12. Firefighting water extraction points

When fighting forest fires, large quantities of extinguishing water must be provided away from supply facilities. The maintenance, expansion, and technical improvement of a network of extinguishing water intake points is therefore necessary, especially in fire-prone areas.

The density and water flow capacity of the network is determined by local needs.

Firefighting water points must be permanently marked and mapped, and access roads must be maintained and signposted.

Firefighting water can be obtained from: **ÿ** natural bodies of water (preferably with access safe for fire engines), **ÿ** cisterns, **ÿ** artificial ponds and wells.

Extraction of extinguishing water using TS

13. Strategies for fighting vegetation fires

Unlike building fires, vegetation firefighting (forest and wildfire fighting) primarily focuses on preventing the fire from spreading. The most important goal is to prevent the fire from spreading from the ground to the treetops and thus triggering a full-blown blaze. In very rare cases, areas can be completely extinguished. The firefighting strategy primarily aims at containing fires.

Hand tools are just as important as extinguishing water in direct and indirect vegetation firefighting and in follow-up firefighting in difficult terrain and far from paved paths. The use of hand tools is still applicable for flames of

ÿ **up to 1 m - fire intensity is low** - hand tools, e.g. fire beaters, can be used in direct attack to control the fire

ÿ **1-3 m - fire is too intense**

- Hand tools cannot be used for direct attacks. Firefighting with Hose lines and backpack sprayers. Cutting clearings with bulldozers. Only flank/parallel attack, depending on the local flame length

ÿ **3m and larger - fire intensity high – extreme**

- extinguishing with hand-held devices is no longer possible. Firefighting with hose lines, tank fire engines and helicopters. Cutting firebreaks with heavy clearing equipment. Because the flames are too high,

  A direct attack is not possible. Due to the excessive risk to emergency personnel, only indirect firefighting is logical. This poses the risk of a full-blown blaze and thus also an increased risk of fire islands caused by flying sparks. Rising hot air can carry sparks from cones, moss, charcoal, or birch bark high into the air and then ignite new fires up to 400 meters ahead of the fire front. This means that flying sparks can even leap over wide strips of hardwood.

13.1 Exploration

To successfully combat a vegetation fire, a thorough reconnaissance of the situation is essential. The operational priorities are set according to the tactical priorities as follows:

1. Protection of people
2. Protection of animals
3. Protection of structures (buildings, roads, utility lines)
4. Protection of vegetation at risk of fire or burning rapidly

Furthermore, the type of fire (full-scale, ground, or area fire), as well as the arrangement and quantity of available fuel, the geographical and topographical features, the main direction of fire spread, characteristics of the terrain, access and escape routes for firefighting vehicles, wind direction, and any expected changes play an important role in the investigation. Conclusions about the fire can also be drawn from observing the smoke column (even when approaching the scene). The color and shape of the smoke column provide information about the fire's behavior. All of this information is used during operational planning to determine how the fire will be fought. Direct or indirect firefighting approaches are possible.

13.2 Locating emergency sites in unmarked areas here

- Signage/marking for emergency vehicles - Vegetation and forest fires are classic emergency scenarios in which reaching the scene is difficult due to a lack of orientation options such as road signs, clear designations, etc. This poses significant problems, particularly for non-local or following emergency personnel.

It is therefore all the more important that the first emergency responders to arrive – be it the fire department or the police – mark the approach to the scene. This is usually done by signalmen. However, since necessary emergency personnel may be unavoidable in an emergency, this is often neglected, which in turn means that subsequent emergency responders can only be guided to the scene through intensive radio communications.

This is where the pilot and reconnaissance team comes in, with the option presented here for marking access routes in unmarked areas. Since each vehicle in the pilot and reconnaissance team carries traffic cones (traffic cones in accordance with BAST TL traffic cones), these can be used to mark the access routes in a simple and effective way.

For this purpose, traffic cones are placed at intersections and turned over on the correct path for further travel so that the tip – similar to an arrow – points in the correct direction of travel.

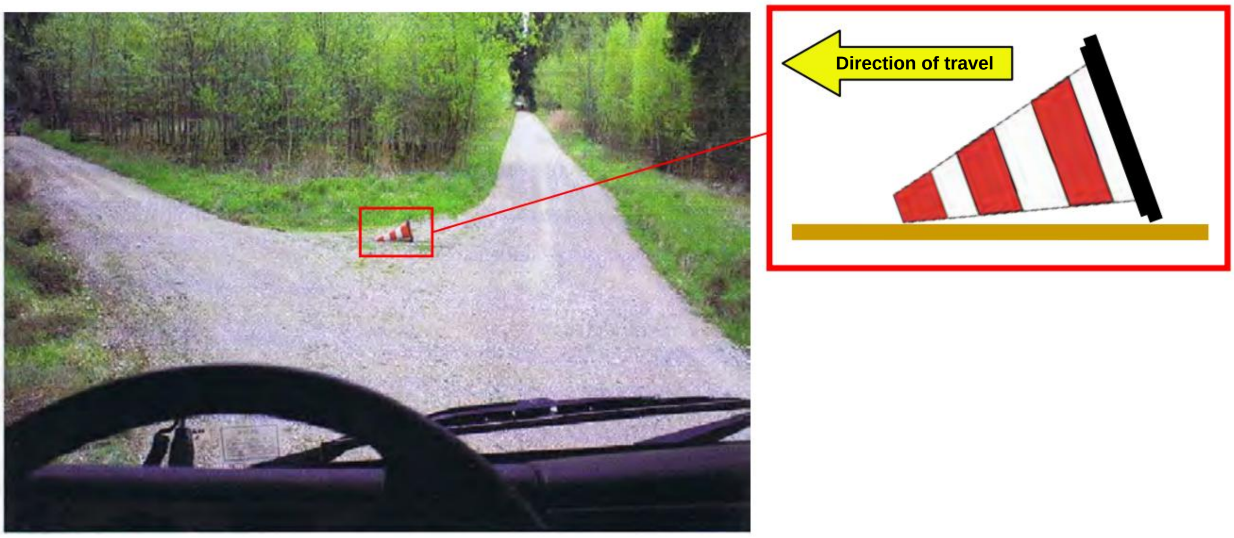


Image: View from the driver's cab - clearly visible for operators and unit commanders, the folded traffic cone points in the direction to be driven.

This simple route marking makes it possible to mark the correct approach for following emergency personnel. With appropriate information, this regulation can be communicated to supporting farmers or other emergency personnel so that they can also follow this marking in the event of an emergency.

Furthermore, this marking can be used not only to indicate the direct access to the scene of the incident, but also to indicate any necessary one-way traffic, e.g. for shuttle traffic to the fire water supply.

14. Firefighting tactics

14.1 Direct vegetation fire fighting

Common practice in firefighting is to directly attack the fire front using firefighting crews, fire engines, and aircraft and/or helicopters with water drops. Fighting against the wind, toward the head of the fire, achieves the greatest effectiveness, but is not without risk due to the difficulty of estimating the rate of spread. This method can only be used for low flame heights. The risk to the firefighters is high if the rate of spread and the length of the flames are misjudged. Rising winds and unpredictable terrain can lead to emergency personnel becoming trapped by the fire. Furthermore, firefighting crews are exposed to the heat of the fire and smoke.

For an example sketch of the setup of a direct firefighting system using the equipment from the vegetation fire modules provided in the Aschaffenburg district, see Appendix 1.

14.2 Indirect vegetation fire fighting

If a direct attack is not possible because the flames are too high, the fire is fought indirectly. The aim is to contain the fire by creating firebreaks or using existing fireproof barriers (roads, paths).

Firebreaks can be created early on and allow for straightened containment lines. Besides the positive effect of being able to work without heat and smoke exposure, this method also has disadvantages. These include increased workload and the danger to emergency personnel working without visual contact with the fire edge. Furthermore, fires are quite capable of destroying these

to jump over firebreaks and monitor the areas behind them
Areas ahead. Important in the control (both direct and indirect
The first step (procedure) is to observe the surroundings in order to immediately extinguish fire islands caused by flying sparks and flying sparks.

For an example sketch of the setup of an indirect firefighting system using the equipment from the vegetation fire modules provided in the Aschaffenburg district, see Appendix 2.

Machine Translated by Google

14.3 Creating and securing firebreaks

Creating firebreaks is generally only useful for containing intense, full-blown fires. The immense human and material resources required for this (clearing equipment, personnel, etc.) are in most cases more effectively utilized in an offensive approach. Nevertheless, creating one can always be considered a plan B to ensure the success of the direct approach. Prior to creation, coordination with the responsible forestry officials and forest owners is essential. The firebreak is always created in two parts. It has a vegetation-free scar and a monitoring area behind it. The scar/firebreak should be laid out so that its width is at least twice the height of the existing vegetation or the length of the flames. Creating firebreaks usually requires the use of large equipment. Forestry and agricultural machinery (Figures A and B), as well as bulldozers (Figure C) or armored recovery vehicles (Figures D and E) of the German Armed Forces can be used to support the soil destruction (exposing the mineral soil).

Image A

Image B

Image C

Images D and E

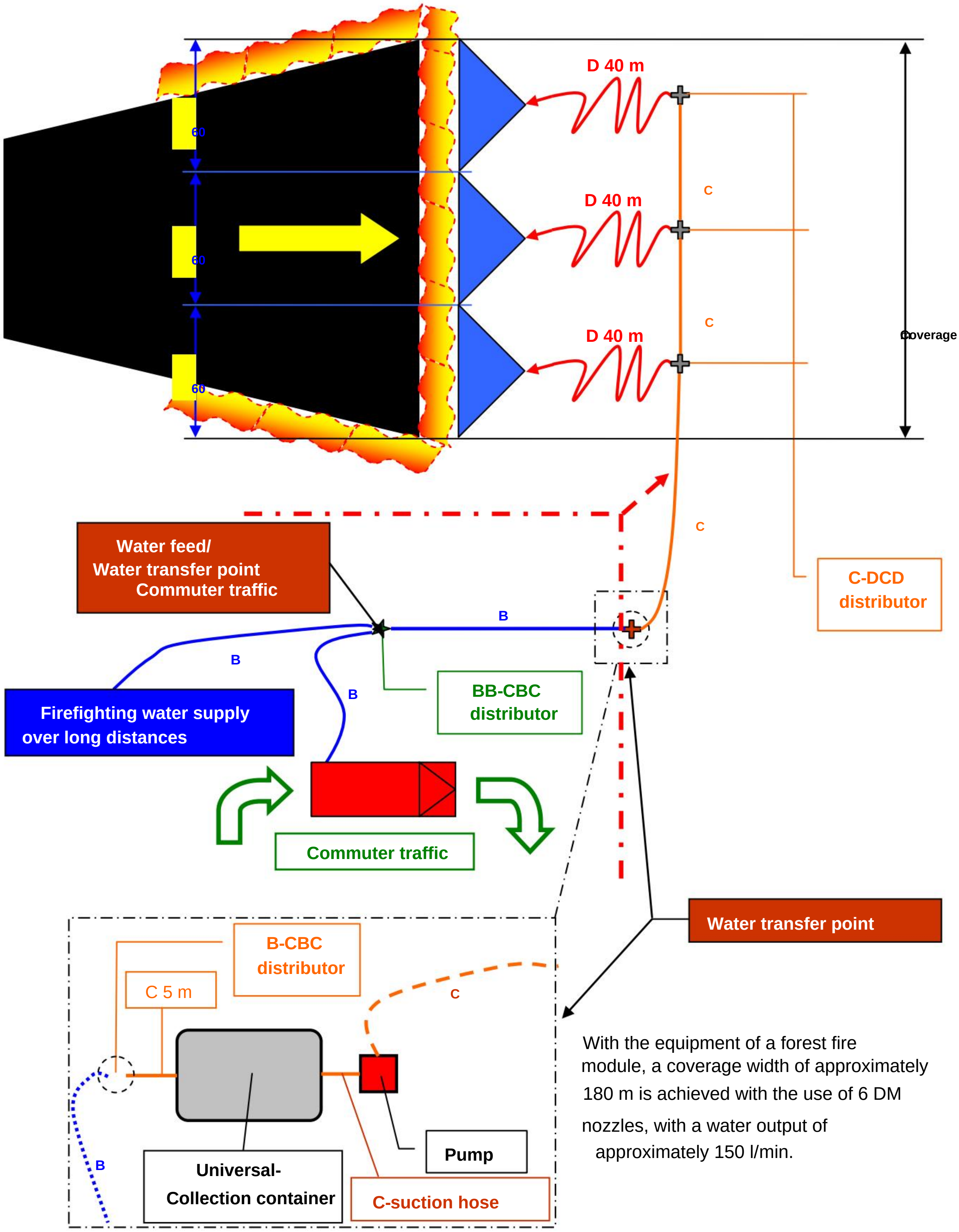
The creation of the lane/the wound strip can also be carried out by fire-fighting teams using hand tools in a predetermined order. The order is determined by the vegetation and soil. Typically, the cutting tools (chainsaw, axe) are used first, followed by hoes (Pulaski, Gorgui multi-tool, flat hoe), rakes (Gorgui multi-tool, McLeod), and shovels in various combinations.

If conditions permit and training is available, the side of the firebreak/slash strip facing the fire is set ablaze to widen and secure the firebreak (pre-fire). Additionally, the vegetation between the firebreak and the pre-fire can be set ablaze to deprive the firebreak of fuel (counter-fire).

The laying of preliminary fire and counter-fire may only be carried out by specially trained personnel.

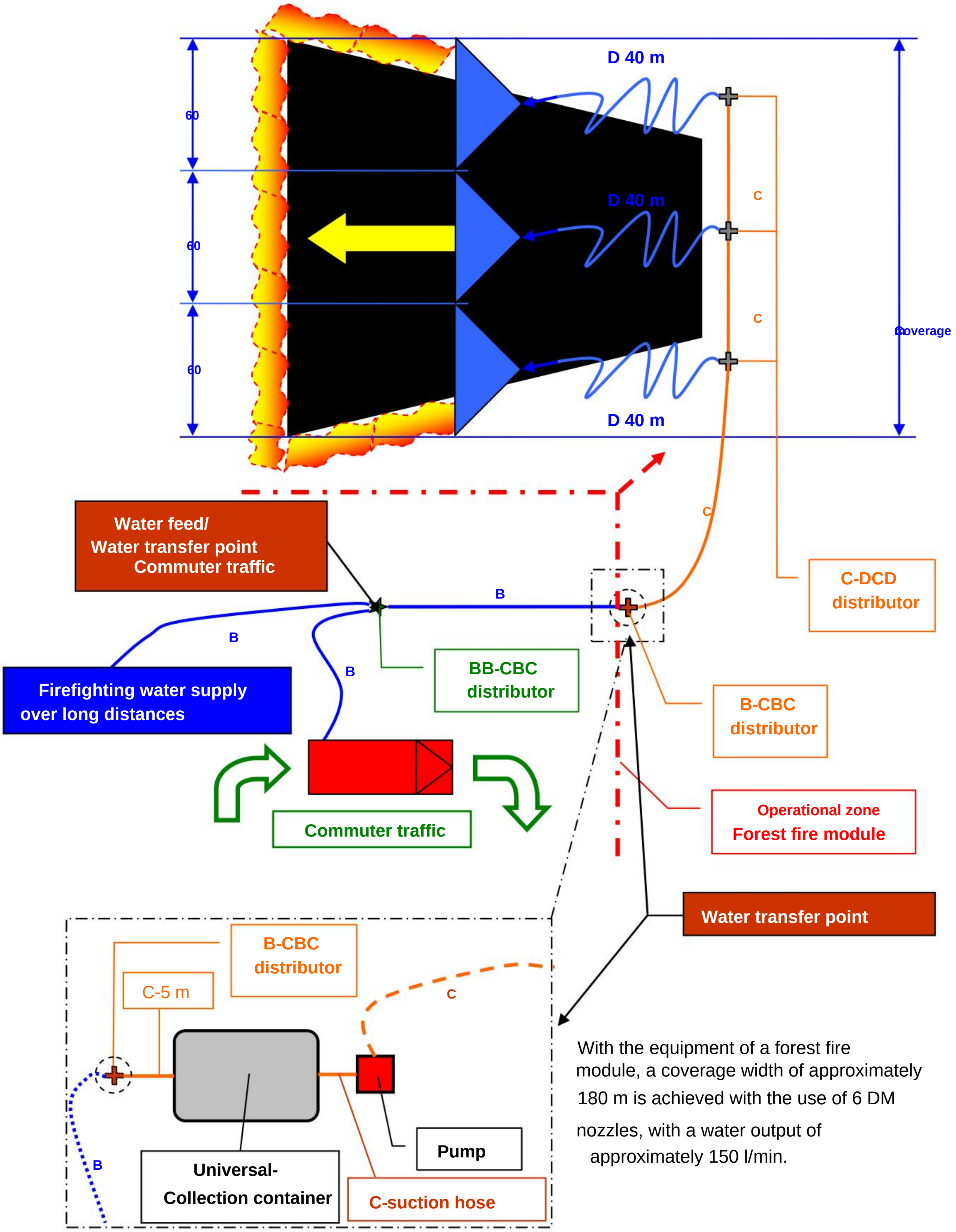
As soon as the edge of the fire has reached the wound strip, crews with water backpacks, fire beaters and shovels secure the firebreak and check the hinterland for fire islands. However, this may only be done if the safety of the emergency services is definitely not at risk. A fully developed forest fire can quickly "overwhelm" a wound strip or firebreak and thus trap emergency services. Sudden winds can also cause a previously "harmless fire" to suddenly assume a very high fire intensity and spread rapidly, ultimately putting emergency services at great risk. For this reason, forest fires should never be fought from the fire front. The basic principle here is: The fire is fought from a safe anchor point (usually the fire's origin) and via the flanks.

APPENDIX 1) **Sketch example – construction of a Direct hole attack (against the direction of fire)**



With the equipment of a forest fire module, a coverage width of approximately 180 m is achieved with the use of 6 DM nozzles, with a water output of approximately 150 l/min.

APPENDIX 2) Sketch example – construction of a Indirect hole attack (in the direction of fire)



With the equipment of a forest fire module, a coverage width of approximately 180 m is achieved with the use of 6 DM nozzles, with a water output of approximately 150 l/min.

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